

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (2 Question × 100 Marks)
CO1 Statement:	<i>Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2</i>		
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Section / Batch:	2023	Date:	

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Decision support system, Operational versus Decision support System, Architecture of data warehouse, ETL, OLAP and data cubes, Dimensional data modelling -star snowflake schemas. Query Processing, Open-source Business Intelligence Tools. Develop an application to perform OLAP operations - Introduction to KNIME tool for reporting.	Q1, Q2	100
GRAND TOTAL:			100 Marks

SECTION A – Technical Presentation

Topic 1: Decision Support Systems (DSS)

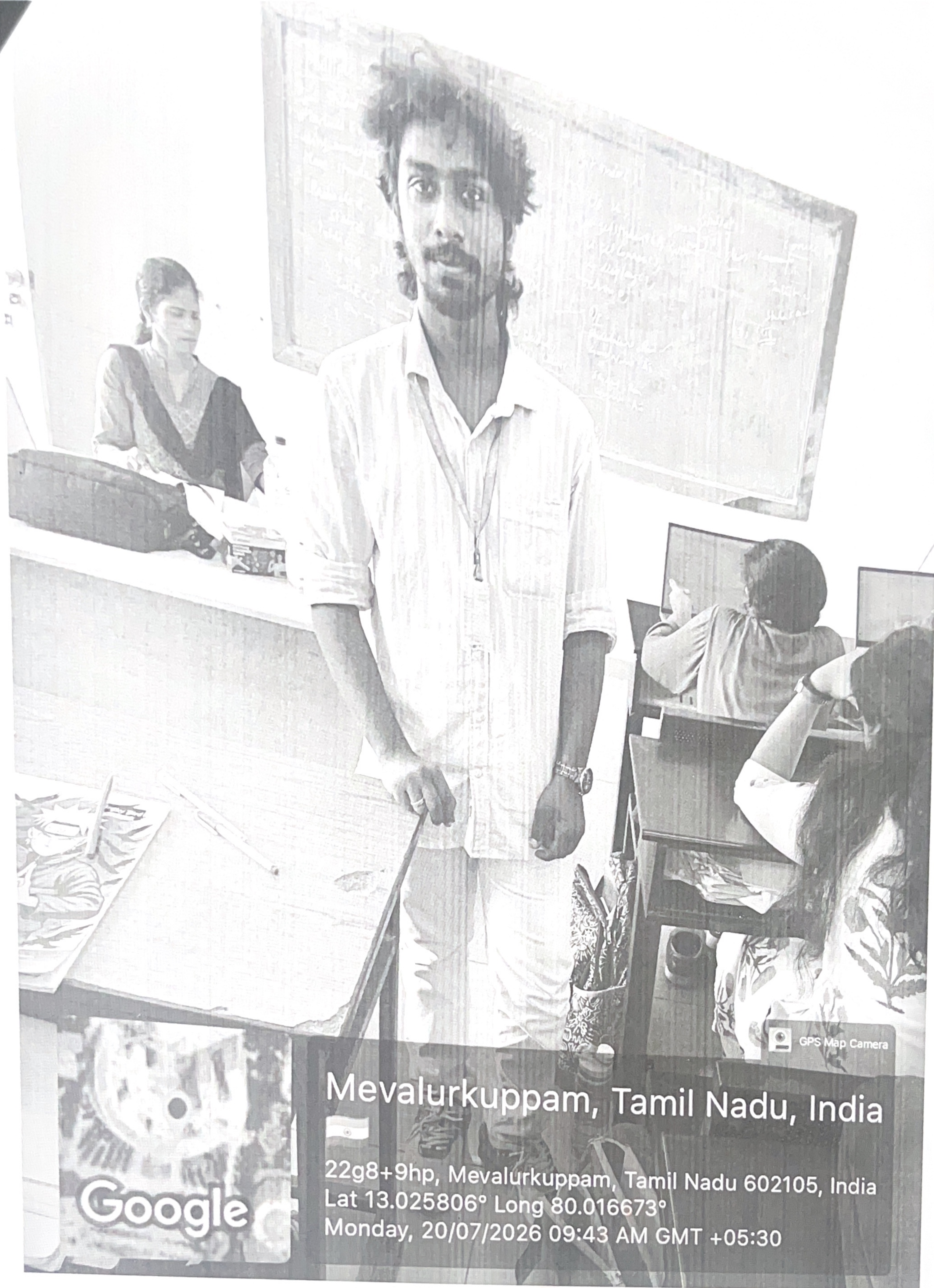
☉ Focus Areas:

- Definition and need of DSS
- Components (Data, Model, UI)
- Types of DSS
- Real-world applications (banking, healthcare)
- Advantages & limitations
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Topic 2: Operational Systems vs DSS

☉ Focus Areas:

- OLTP vs OLAP concepts
- Key differences (data, users, purpose)
- Performance comparison
- Use-case scenarios
- Integration between systems



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OLAP and Data Cubes

A Brief Overview

What is OLAP?

OLAP (Online Analytical Processing) is a computing approach that enables users to analyze large volumes of data from multiple perspectives quickly and interactively. Unlike OLTP (Online Transaction Processing), which focuses on managing day-to-day transactional data, OLAP is designed for complex queries, reporting, and multidimensional analysis used in business intelligence and decision-support systems. OLAP systems allow analysts to perform operations such as slicing, dicing, drilling down, rolling up, and pivoting data to uncover trends, patterns, and insights.

What is a Data Cube?

A data cube is the core data structure used in OLAP systems. It organizes data into multiple dimensions, allowing values (called measures) to be viewed and aggregated along various combinations of dimensions. For example, a sales data cube might have dimensions such as Time, Product, and Location, with a measure like Total Sales. Each cell in the cube represents an aggregated value for a specific combination of dimension members, such as sales of a particular product in a specific region during a specific month.

Key OLAP Operations

- **Roll-up:** Summarizing data by climbing up a concept hierarchy (e.g., city to country).
- **Drill-down:** Breaking down data into finer detail (e.g., year to month).
- **Slice:** Selecting a single dimension value to create a sub-cube.
- **Dice:** Selecting two or more dimensions to create a smaller sub-cube.
- **Pivot (Rotate):** Reorienting the cube to view data from a different perspective.

Types of OLAP Systems

Type	Description
MOLAP	Multidimensional OLAP — stores data in optimized multidimensional array structures (cubes).
ROLAP	Relational OLAP — stores data in relational databases and generates cubes dynamically via SQL.
HOLAP	Hybrid OLAP — combines MOLAP and ROLAP for a balance of performance and scalability.

Applications

OLAP and data cubes are widely used in business intelligence dashboards, financial forecasting, sales and marketing analysis, inventory management, and data warehousing solutions, enabling organizations to make faster, data-driven decisions.